

2023 Bi H2 Q10

Section: Metabolism and Survival

Topic: Genetic Control of Metabolism

Question Summary

The question explores the process of recombinant DNA technology used to genetically modify yeast to produce human transferrin — a protein used to treat anaemia. It focuses on vector choice, enzyme use, and the function of selectable markers.

Worked Solution

(a)(i) The artificial chromosome carrying the human gene into yeast is called a vector. More precisely, it is a yeast artificial chromosome (YAC).

■ Answer: Yeast artificial chromosome (YAC)

(a)(ii) A YAC is used instead of a plasmid because it can:

- Carry larger fragments of DNA (such as whole genes with regulatory regions).
- Ensure proper replication and segregation in yeast cells during cell division.

■ Answer: YACs carry larger DNA fragments and replicate like natural chromosomes.

(b) The same restriction endonuclease is used in stages 1 and 2 to:

- Create complementary sticky ends on both the human gene and the vector.
- Allow them to join accurately by base-pairing between matching sticky ends.

■ Answer: To produce matching sticky ends for successful gene insertion.

(c) Enzyme X, which seals the inserted gene into the artificial chromosome, is DNA ligase.

■ Answer: DNA ligase

(d) Only culture P contained yeast mixed with recombinant artificial chromosomes. These artificial chromosomes carry a selectable marker gene that restores the ability to make leucine. When transferred to plates without leucine, only transformed yeast (those with the recombinant chromosome) could grow. Yeast in culture Q lacked this marker, so they could not synthesise leucine and therefore did not grow.

■ Answer: Only transformed yeast cells in culture P contained the marker gene for leucine synthesis. Culture Q lacked this gene and could not grow without leucine.

Revision Tips

- Vector: Carries foreign DNA into host cells.
- YACs: Used for larger DNA inserts; plasmids for smaller ones.
- Restriction enzymes: Cut DNA at specific sequences; same enzyme ensures matching sticky ends.
- DNA ligase: Joins DNA fragments by sealing sugar–phosphate bonds.
- Selectable marker: Confirms successful transformation (e.g. antibiotic resistance or amino acid synthesis).

